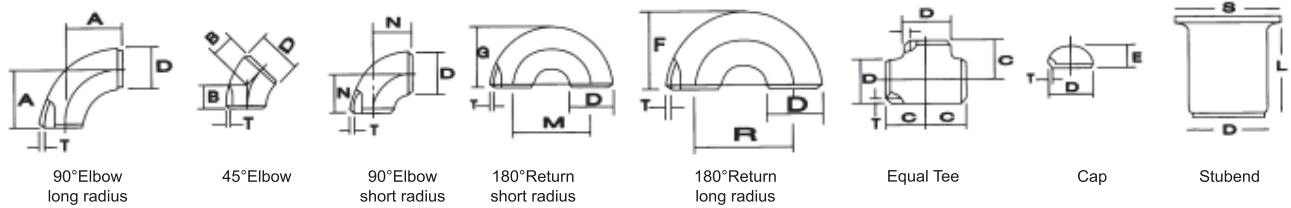


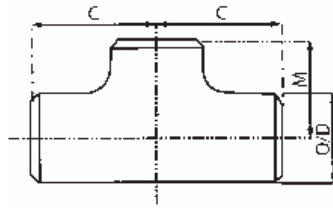
DIMENSION OF BUTT WELD FITTINGS ANSI B-16.9 / B-16.29



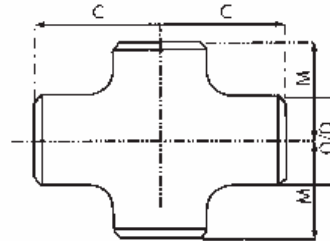
Nominal Pipe Size		Outside Diameter	Center to Face				Back to Face			Center to Center			Length 'L'	
INCH	MM	D	A R=1.5D	B	C	N R=1D	E	F	G	R	M	S	Short L MSSSP43	Long L B16.9
1/2	15	21.3	38.00	16.0	25.0	-	25.0	48.0	-	76.0		35.0	50.8	76.2
3/4	20	26.7	29.00	11.0	29.0	-	25.0	43.0	-	57.0		43.0	50.8	76.2
1	25	33.4	38.00	22.0	38.0	25.0	38.0	56.0	41.0	76.0	51.0	51.0	50.8	101.6
1.1/4	32	42.2	48.00	25.0	48.0	32.0	38.0	70.0	52.0	95.0	64.0	64.0	50.8	101.6
1.1/2	40	48.3	57.15	29.0	57.0	38.0	38.0	83.0	62.0	114.0	76.0	73.0	50.8	101.6
2	50	60.3	76.00	35.0	64.0	51.0	38.0	106.0	81.0	152.0	102.0	93.0	63.5	152.4
2.1/2	65	73.0	95.25	44.0	76.0	64.0	38.0	132.0	100.0	191.0	127.0	105.0	63.5	152.4
3	80	88.9	114.30	51.0	86.0	76.0	51.0	159.0	121.0	229.0	152.0	127.0	63.5	152.4
3.1/2	90	101.6	133.35	57.0	95.0	89.0	64.0	184.0	140.0	267.0	175.0	140.0	76.2	152.4
4	100	114.3	152.0	64.0	105.0	102.0	64.0	210.0	159.0	305.0	203.0	157.0	76.2	152.4
5	125	141.3	190.0	79.0	123.0	127.0	76.0	262.0	197.0	381.0	254.0	186.0	76.2	152.4
6	150	168.3	229.0	95.0	143.0	152.0	89.0	313.0	237.0	457.0	305.0	218.0	88.9	203.2
8	200	219.1	305.0	127.0	178.0	203.0	102.0	414.0	313.0	610.0	406.0	270.0	101.6	203.2
10	250	273.1	381.0	159.0	216.0	254.0	127.0	518.0	391.0	762.0	508.0	324.0	127.0	254.0
12	300	323.8	457.0	190.0	254.0	305.0	152.0	619.0	467.0	914.0	610.0	381.0	152.4	254.0
14	350	355.6	533.0	222.0	279.0	358.0	165.0	711.0	533.0	1067.0	711.0	413.0	152.4	305.0
16	400	406.4	610.0	254.0	305.0	406.0	178.0	813.0	610.0	1219.0	813.0	470.0	152.4	305.0
18	450	457.2	686.0	286.0	343.0	457.0	203.0	914.0	686.0	1372.0	914.0	533.0	152.4	305.0
20	500	508.0	762.0	318.0	381.0	508.0	229.0	1016.0	762.0	1524.0	1016.0	584.0	152.4	305.0
22	550	559.0	838.0	343.0	419.0	559.0	254.0	1118.0	838.0	1676.0	1118.0	614.4	152.4	305.0
24	600	610.0	914.0	381.0	432.0	610.0	267.0	1219.0	914.0	1629.0	1219.0	692.0	152.4	305.0
26	650	660.0	991.0	405.0	495.0	660.0	267.0							
28	700	711.0	1067.0	438.0	521.0	771.0	267.0							
30	750	762.0	1143.0	470.0	559.0	762.0	267.0							
32	800	813.0	1219.0	502.0	597.0	813.0	267.0							
34	850	864.0	1295.0	533.0	635.0	864.0	267.0							
36	900	914.4	1372.0	565.0	673.0	914.0	267.0							

All Dimension in Millimeters

BUTT WELD PIPE FITTINGS AS PER ANSI B-16.9



UNEQUAL TEE

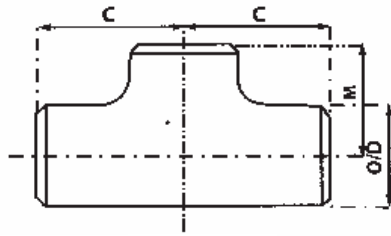


UNEQUAL CROSS

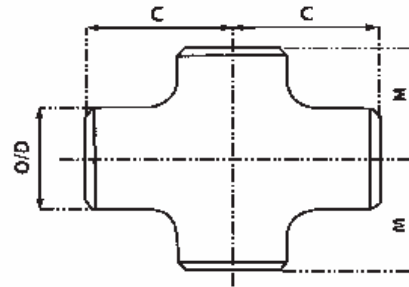
Nominal Pipe Size (NPS)	Outside Diameter at Bevel (O/D)		Center - to - Center	
	Run	Outlet	Run 'C'	Outlet 'M'
1/2" x 3/8"	21.3	17.3	25	25
1/2" x 1/4"	21.3	13.7	25	25
3/4" x 1/2"	26.7	21.3	29	29
3/4" x 3/8"	26.7	17.3	29	29
1" x 3/4"	33.4	26.7	38	38
1" x 1/2"	33.4	21.3	38	38
1 1/4" x 1"	42.2	33.4	48	48
1 1/4" x 3/4"	42.2	26.7	48	48
1 1/4" x 1/2"	42.2	21.3	48	48
1 1/2" x 1 1/4"	48.3	42.2	57	57
1 1/2" x 1"	48.3	33.4	57	57
1 1/2" x 3/4"	48.3	26.7	57	57
1 1/2" x 1/2"	48.3	21.3	57	57
2" x 1 1/2"	60.3	48.3	64	60
2" x 1 1/4"	60.3	42.2	64	57
2" x 1"	60.3	33.4	64	51
2" x 3/4"	60.3	26.7	64	44
2 1/2" x 2"	73.0	60.3	76	70
2 1/2" x 1 1/2"	73.0	48.3	76	67
2 1/2" x 1 1/4"	73.0	42.2	76	64
2 1/2" x 1"	73.0	33.4	76	57
3" x 2 1/2"	88.9	73.0	86	83
3" x 2"	88.9	60.3	86	76
3" x 1 1/2"	88.9	48.3	86	73
3" x 1 1/4"	88.9	42.2	86	70
3 1/2" x 3"	101.6	88.9	95	92
3 1/2" x 2 1/2"	101.6	73.0	95	89
3 1/2" x 2"	101.6	60.3	95	83
3 1/2" x 1 1/2"	101.6	48.3	95	79
4" x 3 1/2"	114.3	101.6	105	102
4" x 3"	114.3	88.9	105	98
4" x 2 1/2"	114.3	73.0	105	95
4" x 2"	114.3	60.3	105	89
4" x 1 1/2"	114.3	48.3	105	86
5" x 4"	141.3	114.3	124	117
5" x 3 1/2"	141.3	101.6	124	114
5" x 3"	141.3	88.9	124	111
5" x 2 1/2"	141.3	73.0	124	108
5" x 2"	141.3	60.3	124	105
6" x 5"	168.3	141.3	143	137
6" x 4"	168.3	114.3	143	130
6" x 3 1/2"	168.3	101.6	143	127
6" x 3"	168.3	88.9	143	124
6" x 2 1/2"	168.3	73.0	143	121
8" x 6"	219.1	168.3	178	168
8" x 5"	218.1	141.3	178	162
8" x 4"	219.1	114.3	178	156
8" x 3 1/2"	219.1	101.6	178	152

Nominal Pipe Size (NPS)	Outside Diameter at Bevel (O/D)		Center - to - Center	
	Run	Outlet	Run 'C'	Outlet 'M'
10" x 8"	273.0	219.1	216	203
10" x 6"	273.0	168.3	216	194
10" x 5"	273.0	141.3	216	191
10" x 4"	273.0	114.3	216	184
12" x 10"	323.8	273.0	254	241
12" x 8"	323.8	219.1	254	229
12" x 6"	323.8	168.3	254	219
12" x 5"	323.8	141.3	254	216
14" x 12"	355.6	323.8	279	270
14" x 10"	355.6	273.0	279	257
14" x 8"	355.6	219.1	279	248
14" x 6"	355.6	168.3	279	238
16" x 14"	406.4	355.6	305	305
16" x 12"	406.4	323.8	305	295
16" x 10"	406.4	273.0	305	283
16" x 8"	406.4	219.1	305	273
16" x 6"	406.4	168.3	305	264
18" x 16"	457.0	406.4	343	330
18" x 14"	457.0	355.6	343	330
18" x 12"	457.0	323.8	343	321
18" x 10"	457.0	273.0	343	308
18" x 8"	457.0	219.1	343	298
20" x 18"	508.0	457.0	381	368
20" x 16"	508.0	406.4	381	356
20" x 14"	508.0	355.6	381	356
20" x 12"	508.0	323.8	381	346
20" x 10"	508.0	273.0	381	333
20" x 8"	508.0	219.1	381	324
22" x 20"	559.0	508.0	419	406
22" x 18"	559.0	457.0	419	394
22" x 16"	559.0	406.4	419	381
22" x 14"	559.0	355.6	419	381
22" x 12"	559.0	323.8	419	371
22" x 10"	559.0	273.0	419	359
24" x 22"	610.0	559.0	432	432
24" x 20"	610.0	508.0	432	432
24" x 18"	610.0	457.0	432	419
24" x 16"	610.0	406.4	432	406
24" x 14"	610.0	355.6	432	406
24" x 12"	610.0	323.8	432	397
24" x 10"	610.0	273.0	432	384
26" x 24"	660.0	610.0	495	483
26" x 22"	660.0	559.0	495	470
26" x 20"	660.0	508.0	495	457
26" x 18"	660.0	457.0	495	444
26" x 16"	660.0	406.4	495	432
26" x 14"	660.0	355.6	495	432
26" x 12"	660.0	323.8	495	422
28" x 26"	711.0	660.0	521	521
28" x 24"	711.0	610.0	521	508
28" x 22"	711.0	559.0	521	495

BUTT WELD PIPE FITTINGS AS PER ANSI B-16.9



EQUAL TEE

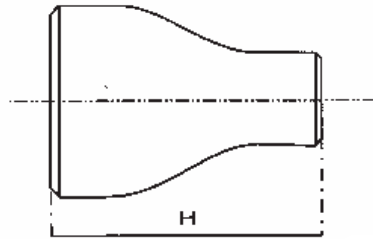


EQUAL CROSS

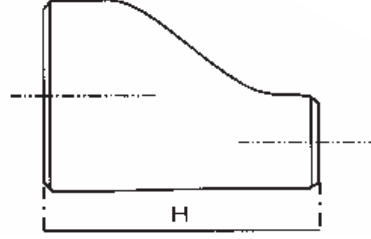
Nominal Pipe Size (NPS)	Outside Diameter at Bevel (O/D)	Run 'C'	Outlet 'M'
1/2"	21.3	25	25
3/4"	26.7	29	29
1"	33.4	38	38
1.1/4"	42.2	48	48
1.1/2"	48.3	57	57
2"	60.3	64	64
2.1/2"	73.0	76	76
3"	88.9	86	86
3.1/2"	101.6	95	95
4"	114.3	105	105
5"	141.3	124	124
6"	168.3	143	143
8"	219.1	178	178
10"	273.0	216	216
12"	323.8	254	254
14"	355.6	279	279
16"	406.4	305	305
18"	457.0	343	343
20"	508.0	381	381
22"	559.0	419	419
24"	610.0	432	432
26"	660.0	495	495
28"	711.0	521	521
30"	762.0	559	559
32"	813.0	597	597
34"	864.0	635	635
36"	914.0	673	673

NOTE : ALL DIMENSIONS ARE IN MM.

BUTT WELD PIPE FITTINGS AS PER ANSI B-16.9



CONC. REDUCER



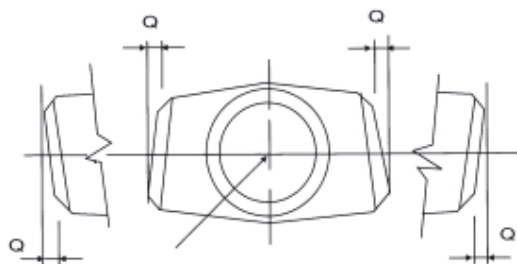
ECC. REDUCER

Nominal Pipe Size (NPS)	Outside Diameter at Bevel (O/D)		End-to-End H
	Run	Outlet	
3/4" x 1/2"	26.7	21.3	38
3/4" x 3/8"	26.7	17.3	38
1" x 3/4"	33.4	26.7	51
1" x 1/2"	33.4	21.3	51
1.1/4" x 1"	42.2	33.4	51
1.1/4" x 3/4"	42.2	26.7	51
1.1/4" x 1/2"	42.2	21.3	51
1.1/2" x 1.1/4"	48.3	42.2	64
1.1/2" x 1"	48.3	33.4	64
1.1/2" x 3/4"	48.3	26.7	64
1.1/2" x 1/2"	46.3	21.3	64
2" x 1.1/2"	60.3	48.3	76
2" x 1.1/4"	60.3	42.2	76
2" x 1"	60.3	33.4	76
2" x 3/4"	60.3	26.7	76
2.1/2" x 2"	73.0	60.3	89
2.1/2" x 1.1/2"	73.0	48.3	89
2.1/2" x 1.1/4"	73.0	42.2	89
2.1/2" x 1"	73.0	33.4	89
3" x 2.1/2"	88.9	73.0	89
3" x 2"	88.9	60.3	89
3" x 1.1/2"	88.9	48.3	89
3" x 1.1/4"	88.9	42.2	89
3.1/2" x 3"	101.6	88.9	102
3.1/2" x 2.1/2"	101.6	73.0	102
3.1/2" x 2"	101.6	60.3	102
3.1/2" x 1.1/2"	101.6	48.3	102
4" x 3.1/2"	114.3	101.6	102
4" x 3"	114.3	88.9	102
4" x 2.1/2"	114.3	73.0	102
4" x 2"	114.3	60.3	102
4" x 1.1/2"	114.3	48.3	102
5" x 4"	141.3	114.3	127
5" x 3.1/2"	141.3	101.6	127
5" x 3"	141.3	88.9	127
5" x 2.1/2"	141.3	73.0	127
5" x 2"	141.3	60.3	127
6" x 5"	168.3	141.3	140
6" x 4"	168.3	114.3	140
6" x 3.1/2"	168.3	101.6	140
6" x 3"	168.3	88.9	140
6" x 2.1/2"	168.3	73.0	140
8" x 6"	219.1	168.3	152
8" x 5"	219.1	141.3	152
8" x 4"	219.1	114.3	152
8" x 3.1/2"	219.1	101.6	152
10" x 8"	273.0	219.1	178
10" x 6"	273.0	168.3	178
10" x 5"	273.0	141.3	178
10" x 4"	273.0	114.3	178

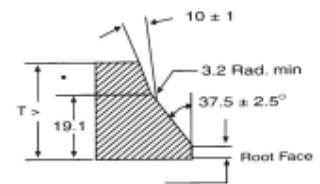
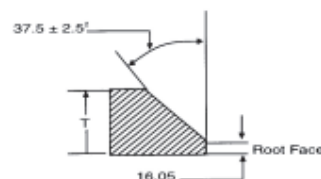
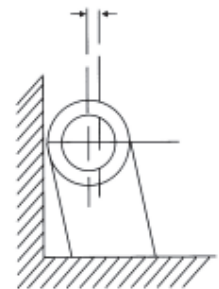
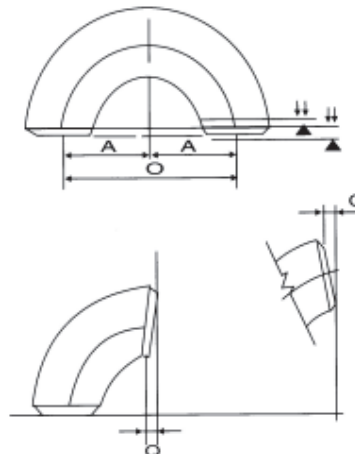
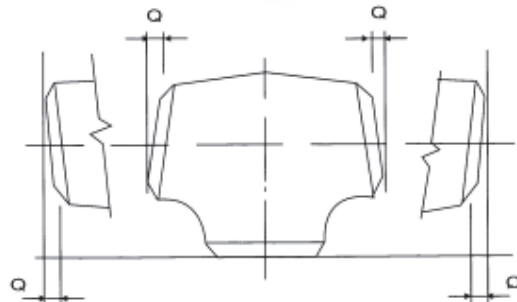
Nominal Pipe Size (NPS)	Outside Diameter at Bevel (O/D)		End-to-End H
	Run	Outlet	
12" x 10"	323.8	273.0	203
12" x 8"	323.8	219.1	203
12" x 6"	323.8	168.3	203
12" x 5"	323.8	141.3	203
14" x 12"	355.6	323.8	330
14" x 10"	355.6	273.0	330
14" x 8"	355.6	219.1	330
14" x 6"	355.6	168.3	330
16" x 14"	406.4	355.6	356
16" x 12"	406.4	323.8	356
16" x 10"	406.4	273.0	356
16" x 8"	406.4	219.1	356
18" x 16"	457.0	406.4	381
18" x 14"	457.0	356.6	381
18" x 12"	457.0	323.8	381
18" x 10"	457.0	273.0	381
20" x 18"	508.0	457.0	508
20" x 16"	508.0	406.4	508
20" x 14"	508.0	355.6	508
20" x 12"	508.0	323.8	508
22" x 20"	559.0	508.0	508
22" x 18"	559.0	457.0	508
22" x 16"	559.0	406.4	508
22" x 14"	559.0	355.6	508
24" x 22"	610.0	559.0	508
24" x 20"	610.0	508.0	508
24" x 18"	610.0	457.0	508
24" x 16"	610.0	406.4	508
26" x 24"	660.0	610.0	610
26" x 22"	660.0	559.0	610
26" x 20"	660.0	508.0	610
26" x 18"	660.0	457.0	610
28" x 26"	711.0	660.0	610
28" x 24"	711.0	610.0	610
28" x 20"	711.0	508.0	610
28" x 18"	711.0	457.0	610
30" x 28"	762.0	711.0	610
30" x 26"	762.0	660.0	610
30" x 24"	762.0	610.0	610
30" x 20"	762.0	508.0	610
32" x 30"	813.0	762.0	610
32" x 28"	813.0	711.0	610
32" x 26"	813.0	660.0	610
32" x 24"	813.0	610.0	610
34" x 32"	864.0	813.0	610
34" x 30"	864.0	762.0	610
34" x 26"	864.0	660.0	610
34" x 24"	864.0	610.0	610

DIMENSION TOLERANCE ANSI B 16.9 / B16.28 / MSS SP - 43

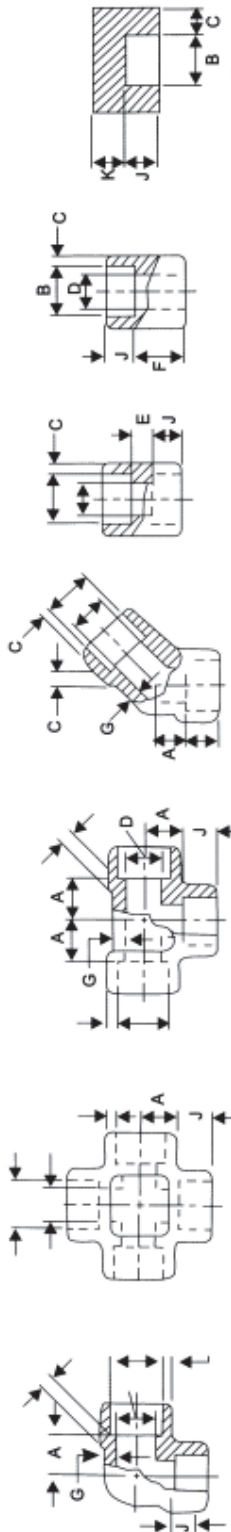
ALL FITTINGS						90°/60°/45° /30° ELBOWS & TEES		REDUCERS		180° RETURNS						CAPS		ANGULARITY TOLERANCE																
Nominal Pipe Size Inch/mm		Outside Diameter at Level		Inside Dia meter	Wall Thickness at End		Centre to End		Overall Length Dimensions		Centre to End		Back to Face Dimensions		Alignment of End Dimensions		Overall length		Nominal Pipe Size	Off Angle Inch/mm		Off Plane												
D				T		A,B,C,M		H		O		K		U		E			Q		P													
	(1) B16.9	MSS SP43	(2) B16.9	B16.9	MSS SP43	B16.9	MSS SP43	B16.9	MSS SP43	B16.9	MSS SP43	B16.9	MSS SP43	B16.9	MSS SP43	B16.9	MSS SP43		B16.9	MSS SP43	B16.9													
1/2" -2 1/2" 15 - 65	±1.6 - 0.8	±0.80	±0.8	Not Less Than 87.5% Nominal Thk.		FROM 1/2" TO 18" 15 TO 600	FROM 3/4"	FROM 1/2"-24" 15 - 600	FROM 1/2"-28" 15 - 200							±3	±3.17	1/2" - 4" 15 - 100	±1	16" - 24" 400 - 600 1.6	±2													
3" - 3 1/2" 80 - 90	±1.6					±1.6	±2	±1.60	±2									±1.60	±6		±6.35	±6.0	±6.4	±1	±0.8	5" - 8" 125 - 200	±2	±4						
4" 100																										10" - 12" 250 - 300	±3	±5						
5" - 6" 125 - 150	±2.4 -1.6	+1.60	±3.2					10" -24" +2.38 250 - 600	±10"	±10"				±2.0"	±1.60	14" - 16" 350 - 400	±3	±7																
8" 200																-0.80	18" - 24" 450 - 600	±3	±10															
10" - 18" 250 - 450	+4 -3.2	+2.38 -0.80	±3.2					±2.40												±7	±6.35	26" - 30" 650 - 750	±4	26" - 36" 650- 900 2.4	±10									
20" - 24" 500 - 600	+6.4 -4.8	3.17 0.79																														32" - 42" 800 - 1050	±5	±13
26"- 30" 650 - 750	+6.4 -4.8	±4.8						±4.8														±3	FROM 26"- 48" 650NB - 701200 NB ±5											
32" -48" 800 - 1200	+6.4 -4.8					±5	42" - 48" 1050 - 1200	±5	±20																									



This End Flush Against Square



DIMENSIONS OF SOCKET-WELD FITTINGS ANSI B 16.11



90°Elbow	Cross			Tee		45° Elbow		Coupling			Half Coupling			Cap															
	Nominal Pipe Size	Socket Bore Diameter B Note 1	Bore Diameter of Fitting D [Note (1)]	Socket Wall Thickness, C [Note (2)]			Body wall G		Min. Depth of Socket J	Center to Bottom of Socket, A			Laying Lengths		Tolerances		End Wall Thickness K Min												
				Class Designation			Class Designation			Class Designation			Couplings E	Half Couplings F	A	E	F	Class Designation											
				3000	6000	9000	3000	6000		9000	3000	6000						9000	3000	6000	9000								
Avg. Min.																													
6	1/8	11.2	7.6	4.8	...	3.18	3.18	3.96	3.43	...	2.41	3.15	...	9.5	11.0	11.0	...	8.0	8.0	...	6.5	16.0	1.0	1.5	1.0	4.8	6.4	...	
		10.8	6.1	3.2	...	...	...	...	...	...	...	...	...	...	9.5	11.0	13.5	...	8.0	8.0	...	6.5	16.0	1.0	1.5	1.0	4.8	6.4	...
		14.6	10.0	7.1	...	3.78	3.30	4.60	4.01	...	3.02	3.68	...	9.5	11.0	13.5	...	8.0	8.0	...	6.5	16.0	1.0	1.5	1.0	4.8	6.4	...	
8	1/4	14.2	8.5	5.6	...	...	...	...	...	...	3.20	4.01	...	9.5	13.5	15.5	...	8.0	11.0	...	6.5	17.5	1.5	3.0	1.5	4.8	6.4	...	
		18.0	13.3	9.9	...	4.01	3.50	5.03	4.37	...	3.20	4.01	...	9.5	13.5	15.5	...	8.0	11.0	...	6.5	17.5	1.5	3.0	1.5	4.8	6.4	...	
10	3/8	17.6	11.8	8.4	...	...	...	...	...	...	3.20	4.01	...	9.5	13.5	15.5	...	8.0	11.0	...	6.5	17.5	1.5	3.0	1.5	4.8	6.4	...	
15	1/2	22.2	16.6	12.5	7.2	4.67	4.09	5.97	5.18	9.35	8.18	3.73	4.78	7.47	9.5	15.5	19.5	25.5	11.0	12.5	15.5	9.5	22.5	1.5	3.0	1.5	6.4	7.9	11.2
		21.8	15.0	11.0	5.6	...	...	...	...	...	...	3.73	4.78	7.47	9.5	15.5	19.5	25.5	11.0	12.5	15.5	9.5	22.5	1.5	3.0	1.5	6.4	7.9	11.2
20	3/4	27.6	21.7	16.3	11.8	4.90	4.27	6.96	6.04	9.78	8.56	3.91	5.56	7.82	12.5	19.0	22.5	28.5	13.0	14.0	19.0	9.5	24.0	1.5	3.0	1.5	6.4	7.9	12.7
		27.2	20.2	14.8	10.3	...	...	...	...	...	...	3.91	5.56	7.82	12.5	19.0	22.5	28.5	13.0	14.0	19.0	9.5	24.0	1.5	3.0	1.5	6.4	7.9	12.7
25	1	34.3	27.4	21.5	16.0	5.69	4.98	7.92	6.93	11.38	9.96	4.55	6.35	9.09	12.5	22.5	27.0	32.0	14.0	17.5	20.5	12.5	28.5	2.0	4.0	2.0	9.6	11.2	14.2
		33.9	25.9	19.9	14.4	...	...	...	...	...	...	4.55	6.35	9.09	12.5	22.5	27.0	32.0	14.0	17.5	20.5	12.5	28.5	2.0	4.0	2.0	9.6	11.2	14.2
32	1 1/4	43.1	35.8	30.2	23.5	6.07	5.28	7.92	6.93	12.14	10.62	4.85	6.35	9.70	12.5	27.0	32.0	35.0	17.5	20.5	22.5	12.5	30.0	2.0	4.0	2.0	9.6	11.2	14.2
		42.7	34.3	28.7	22.0	...	...	...	...	...	...	4.85	6.35	9.70	12.5	27.0	32.0	35.0	17.5	20.5	22.5	12.5	30.0	2.0	4.0	2.0	9.6	11.2	14.2
40	1 1/2	49.2	41.6	34.7	28.7	6.35	5.54	8.92	7.80	12.70	11.12	5.08	7.14	10.15	12.5	32.0	38.0	38.0	20.5	25.5	25.5	12.5	32.0	2.0	4.0	2.0	11.2	12.7	15.7
		48.8	40.1	33.2	27.2	...	...	...	...	...	...	5.08	7.14	10.15	12.5	32.0	38.0	38.0	20.5	25.5	25.5	12.5	32.0	2.0	4.0	2.0	11.2	12.7	15.7
50	2	61.7	53.3	43.6	38.9	6.93	6.04	10.92	9.50	13.84	12.12	5.54	8.74	11.07	16.0	38.0	41.0	54.0	25.5	28.5	28.5	19.0	41.0	2.0	4.0	2.0	12.7	15.7	19.0
		61.2	51.7	42.1	37.4	...	...	...	...	...	...	5.54	8.74	11.07	16.0	38.0	41.0	54.0	25.5	28.5	28.5	19.0	41.0	2.0	4.0	2.0	12.7	15.7	19.0
65	2 1/2	74.4	64.2	...	...	8.76	7.67	...	...	...	...	7.01	...	...	16.0	41.0	...	...	28.5	...	...	19.0	43.0	2.5	5.0	2.5	15.7	19.0	...
		73.9	61.2	...	...	...	...	...	...	...	...	7.01	...	...	16.0	41.0	...	...	28.5	...	...	19.0	43.0	2.5	5.0	2.5	15.7	19.0	...
80	3	90.3	79.4	...	...	9.52	8.30	...	...	...	...	7.62	...	...	16.0	57.0	...	...	32.0	...	...	19.0	44.5	2.5	5.0	2.5	19.0	22.4	...
		89.8	76.4	...	...	...	...	...	...	...	...	7.62	...	...	16.0	57.0	...	...	32.0	...	...	19.0	44.5	2.5	5.0	2.5	19.0	22.4	...
100	4	115.7	103.8	...	...	10.69	9.35	...	...	...	...	8.56	...	...	19.0	66.5	...	...	41.0	...	...	19.0	48.0	2.5	5.0	2.5	22.4	28.4	...
		115.2	100.7	...	...	...	...	...	...	...	...	8.56	...	...	19.0	66.5	...	...	41.0	...	...	19.0	48.0	2.5	5.0	2.5	22.4	28.4	...

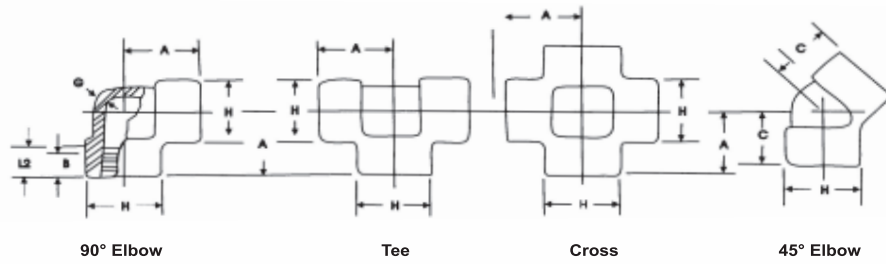
Avg. Min.

(1) All Dimensions Are In Millimeters

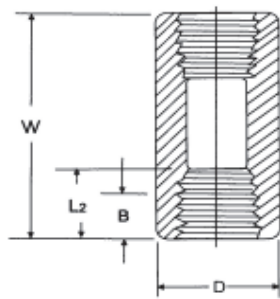
(2) Upper and lower values for each size are the respective maximum and minimum dimensions.

(3) Average of socket wall thickness around periphery shall be no less than listed values. The minimum values are permitted in localized areas.

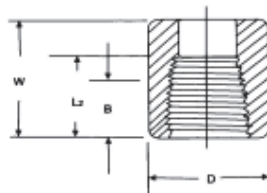
DIMENSIONS OF THREADED FITTINGS ANSI B 16.11



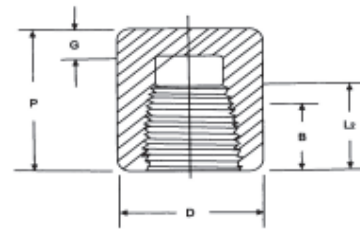
Nominal Pipe Size		Center to End Elbows Tees, Crosses A			Center to End 45 Elbows C			Outside Diameter of Bend H			Minimum Wall Thickness G			Min. Length of Thread Note - 1	
NB	INCH	2000	3000	6000	2000	3000	6000	2000	3000	6000	2000	3000	6000	B	L2
6	1/8	21	21	25	17	17	19	22	22	25	3.18	3.18	6.35	6.4	6.7
8	1/4	21	25	28	17	19	22	22	25	33	3.18	3.30	6.60	8.1	10.2
10	3/8	25	28	33	19	22	25	25	33	38	3.18	3.51	6.98	9.1	10.4
15	1/2	28	33	38	22	25	28	33	38	46	3.18	4.09	8.15	10.9	13.6
20	3/4	33	38	44	25	28	33	38	46	56	3.18	4.32	8.53	12.7	13.9
25	1	38	44	51	28	33	35	46	56	62	3.68	4.98	9.93	14.7	17.3
32	1 1/4	44	51	60	33	35	43	56	62	75	3.89	5.28	10.59	17.0	18.0
40	1 1/2	51	60	64	35	43	44	62	75	84	4.01	5.56	11.07	17.8	18.4
50	2	60	64	83	43	44	52	75	84	102	4.27	7.14	12.09	19.0	19.2
65	2 1/2	76	83	95	52	52	64	92	102	121	5.61	7.65	15.29	23.6	28.9
80	3	86	95	106	64	64	79	109	121	146	5.99	8.84	16.64	25.9	30.5
100	4	106	114	114	79	79	79	146	152	152	6.55	11.18	18.67	27.7	33.0



Coupling



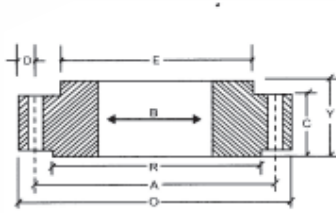
Half-Coupling



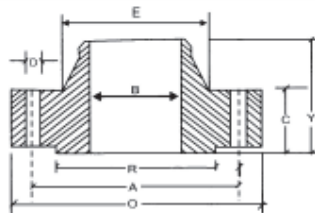
Cap

Nominal Pipe Size		End to End Couplings W	End to End Caps P		Outside Diameter D		End Wall Thickness min. G		Min. Length of Thread Note - 1	
NB	INCH	3000 and 6000	3000	6000	3000	6000	3000	6000	B	L2
6	1/8	32	19	-	16	22	4.8	-	6.4	6.7
8	1/4	35	25	27	19	25	4.8	6.4	8.1	10.2
10	3/8	38	25	27	22	32	4.8	6.4	9.1	10.4
15	1/2	48	32	33	28	38	6.4	7.9	10.9	13.6
20	3/4	51	37	38	35	44	6.4	7.9	12.7	13.9
25	1	60	41	43	44	57	9.7	11.2	14.7	17.3
32	1 1/4	67	44	46	57	64	9.7	11.2	17.0	18.0
40	1 1/2	79	44	48	64	76	11.2	12.7	17.8	18.4
50	2	86	48	51	76	92	12.7	15.7	19.0	19.2
65	2 1/2	92	60	64	92	108	15.7	19.0	23.6	28.9
80	3	108	65	68	108	127	19.0	22.4	25.9	30.5
100	4	121	68	75	140	159	22.4	28.4	27.7	33.0

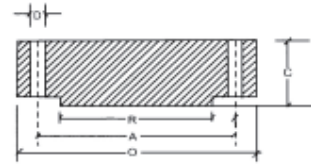
DIMENSIONS OF FORGED FLANGES ANSI B 16.5



SLIP-ON FLANGE



WELDING NECK FLANGE



BLIND FLANGE

ASA 150 CLASS

Nominal Pipe Size		Flange Dia	Dia of Bolt Circle	Dia of Bolt Holes	No. of Holes	Thk of Flange	Dia of Hub	Length through Hub			Dia of Bore		Dia of R/F	Depth of Socket	Pipe Dia
								S/O & S/W	W/N	L/J	S/O & S/W	L/J			
(MM)	(INCH.)	O	A	D		C	E	Y	Y	Y	B	B	R	F	X
15	1/2	90.0	60.3	15.9	4	11.1	30.2	15.9	47.6	15.9	22.3	22.9	34.9	9.5	21.33
20	3/4	100.0	69.8	15.9	4	12.7	38.1	15.9	52.4	15.9	27.7	28.2	42.9	11.1	26.67
25	1	110.0	79.4	15.9	4	14.3	49.2	17.5	55.6	17.5	34.5	35.0	50.8	12.7	33.40
32	1 1/4	115.0	88.9	15.9	4	15.9	58.7	20.6	57.1	20.6	43.2	43.7	63.5	14.3	42.16
40	1 1/2	125.0	98.4	15.9	4	17.5	65.1	22.2	61.9	22.2	49.5	50.0	73.0	15.9	48.26
50	2	150.0	120.6	19.0	4	19.0	77.8	25.4	63.5	25.4	62.0	62.5	92.1	17.5	60.31
65	2 1/2	180.0	139.7	19.0	4	22.2	90.5	28.6	69.8	28.6	74.7	75.4	104.8	19.0	73.02
80	3	190.0	152.4	19.0	4	23.8	107.9	30.2	69.8	30.2	90.7	91.4	127.0	20.6	88.90
100	4	230.0	190.5	19.0	8	23.8	134.9	33.3	76.2	33.3	116.1	116.8	157.2	23.8	114.30
125	5	255.0	215.9	22.2	8	23.8	163.5	36.5	88.9	36.5	143.8	144.5	185.7	23.8	141.30
150	6	280.0	241.3	22.2	8	25.4	192.1	39.7	88.9	39.7	170.7	171.4	215.9	27.0	168.27
200	8	345.0	298.4	22.2	8	28.6	246.1	44.4	101.6	44.4	221.5	222.2	269.9	31.7	219.07
250	10	405.0	361.9	25.4	12	30.2	304.8	49.2	101.6	49.2	276.3	277.4	323.8	33.3	273.05
300	12	485.0	431.8	25.4	12	31.8	365.1	55.6	114.3	55.6	327.1	328.2	381.0	39.7	323.85
350	14	535.0	476.2	28.6	12	34.9	400.0	57.1	127.0	79.4	359.1	360.2	412.7	41.3	355.60
400	16	595.0	539.7	28.6	16	36.5	457.2	63.5	127.0	87.3	410.5	411.2	469.9	44.4	406.40
450	18	635.0	577.8	31.7	16	39.7	504.8	68.3	139.7	96.8	461.8	462.3	533.4	49.2	457.20
500	20	700.0	635.0	31.7	20	42.9	558.8	73.0	144.5	103.2	513.1	514.3	584.2	54.0	508.00
600	24	815.0	749.3	34.9	20	47.6	663.6	82.5	152.4	111.1	615.9	615.9	692.1	63.5	609.60

All Dimensions are in Millimeters ● Flanges except Lap Joint will be furnished with (1.6mm) Raised Face, which is included in Thickness(C) and Length through Hub(Y).

ASA 300 CLASS

Nominal Pipe Size		Flange Dia	Dia of Bolt Circle	Dia of Bolt Holes	No. of Holes	Thk of Flange	Dia of Hub	Length through Hub			Dia of Bore		Dia of R/F	Depth of Socket	Pipe Dia
								S/O & S/W	W/N	L/J	S/O & S/W	L/J			
(MM)	(INCH.)	O	A	D		C	E	Y	Y	Y	B	B	R	F	X
15	1/2	95.0	66.7	15.9	4	14.3	38.1	22.2	52.4	22.2	22.3	22.9	34.9	9.5	21.33
20	3/4	115.0	82.5	19.0	4	15.9	47.6	25.4	57.1	25.4	27.7	28.2	42.9	11.1	26.67
25	1	125.0	88.9	19.0	4	17.5	54.0	27.0	61.9	27.0	34.5	35.0	50.8	12.7	33.40
32	1 1/4	135.0	98.4	19.0	4	19.0	63.5	27.0	65.1	27.0	43.2	43.7	63.5	14.3	42.16
40	1 1/2	155.0	114.3	22.2	4	20.6	69.8	30.2	68.3	30.2	49.5	50.0	73.0	15.9	48.26
50	2	165.0	127.0	19.0	8	22.2	84.1	33.3	69.8	33.3	62.0	62.5	92.1	17.5	60.31
65	2 1/2	190.0	149.2	22.2	8	25.4	100.0	38.1	76.2	38.1	74.7	75.4	104.8	19.0	73.02
80	3	210.0	168.3	22.2	8	28.6	117.5	42.9	79.4	42.9	90.7	91.4	127.0	20.6	88.90
100	4	255.0	200.0	22.2	8	31.8	146.0	47.6	85.7	47.6	116.1	116.8	157.2	23.8	114.30
125	5	280.0	234.9	22.2	8	34.9	177.8	50.8	98.4	50.8	143.8	144.5	185.7	-	141.30
150	6	320.0	269.9	22.2	12	36.5	206.4	52.4	98.4	52.4	170.7	171.4	215.9	-	168.27
200	8	380.0	330.2	25.4	12	41.3	260.3	61.9	111.1	61.9	221.5	222.2	269.9	-	219.07
250	10	445.0	387.3	28.6	16	47.6	320.7	66.7	117.5	95.2	276.3	277.4	323.8	-	273.05
300	12	520.0	450.8	31.7	16	50.8	374.6	73.0	130.2	101.6	327.1	328.2	381.0	-	323.85
350	14	585.0	514.3	31.7	20	54.0	425.4	76.2	142.9	111.1	359.1	360.2	412.7	-	355.60
400	16	650.0	571.5	34.9	20	57.2	482.6	82.5	146.0	120.6	410.5	411.2	469.9	-	406.40
450	18	710.0	628.5	34.9	24	60.3	533.4	88.9	158.7	130.2	461.8	462.3	533.4	-	457.20
500	20	775.0	685.8	34.9	24	63.5	587.4	95.2	161.9	139.7	513.1	514.3	584.2	-	508.00
600	24	915.0	812.8	41.3	24	69.8	701.7	106.4	168.3	152.4	615.9	615.9	692.1	-	609.60

All Dimensions are in Millimeters ● Flanges except Lap Joint will be furnished with (1.6mm) Raised Face, which is included in Thickness(C) and Length through Hub(Y).